# Multilabel Classification for Cervical Cancer Diagnosis: a comparative approach based on different imputation methods

Filippo Bonaldo
*Alma Mater Studiorum*
*Digital Transformation Management*
*Machine Learning and Data Mining*

***Abstract*—Cervical cancer is a largely preventable disease whose slow progression offers a critical window for early detection via established screening protocols. The application of Machine Learning techniques to this domain could significantly improve efficiency in the related screening and prevention methods. This study conducts a Multilabel Classification task in order to predict the outcome of the four diagnostic tests: Hinselmann, Schiller, Cytology, and Biopsy. However, clinical datasets of this kind are commonly affected by high-missing data rates and severe class imbalance, therefore in this work it is also offered a comparative view on how different data preprocessing procedures yield different outcomes.**

## I. Introduction

This study aims to demonstrate how Machine Learning (ML) techniques can be applied in this context, specifically that of cervical cancer screening.

Cervical cancer develops in the cervix, the lower, narrow end of the uterus that connects to the vagina. Unlike many other cancers, cervical cancer is almost entirely preventable and has a very well-understood primary cause. The Human Papillomavirus (HPV), the most common sexually transmitted infection globally, is widely recognized as the primary catalyst for over 95% of cervical cancer cases. Despite this, only a small fraction of the substantial number of women contracting HPV develop cancer, but research reveals that having a history of multiple different sexually transmitted diseases (STDs) significantly weakens the local immune response in the cervical tissue, making it easier for the HPV virus to persist and cause cellular mutation in the long run. In fact, cervical cancer typically takes 10 to 20 years to manifest, and it is this very slow progression that underlines the extreme importance of early screening: if doctors are able to "intercept" the development of cancer at an early stage, they can effectively remove the mutated cells before they become malignant.

Four types of screening tests are generally employed in this scenario, and each of them is associated with varying levels of severity. The Cytology test is always the first one to be performed, followed by the Hinselmann and the Schiller's tests (usually performed simultaneously). The Biopsy is the definitive diagnostic test to carry out and, due to its invasiveness, patient discomfort, and high cost, it is performed only if the previous two diagnostic tests reveal suspicious patterns. Multilabel Classification is an appropriate method in this context: the objective is not merely to predict whether a Biopsy test's result is positive or not, but also to map a precise risk profile for a patient, considering her history and the outcomes of the diagnostic tests. In this sense, Data Preprocessing played a relevant role, refining the data and applying feature transformation to create a clearer and more significant mapping, to be then fed to the chosen classifiers. Specifically, a diverse suite of classifiers was chosen to accomplish the intended task: the linear Logistic Regression, the tree-based Random Forest and its gradient-boosted variant XGBoost, the Support Vector Machine's Classifier (referred to as SVC), and the "lazy learner" KNN algorithm. All these algorithms had to execute a multilabel classification task, where each of them has to predict the value (binary flag, with "0" implying negativity, while "1" positivity) for each of the target variables simultaneously. The goal is to understand not just which algorithm performed best, but also how different choices made during Data Preparation concretely impact the classification process.

The remainder of this work details the actual process, where each phase's output becomes the input for the next one. Data exploration allowed for a deep understanding of the raw data at hand, while Data Preprocessing manipulated the data so to make it suitable for classification. The modeling phase employed the aforementioned classifiers to make predictions and extract relevant information from the cleaned data, ultimately extracting actionable insights regarding both cervical cancer diagnosis and the broader integration of Machine Learning into the medical screening ecosystem.

## II. Proposed Method

### A. Exploratory Data Analysis

This study utilizes the "Cervical Cancer (Risk Factors)" dataset from the UCI Machine Learning Repository, comprising 858 clinical records from the Hospital of Caracas, Venezuela. The 36 variables characterizing each entry include the 4 target variables (i.e., the four diagnostic tests) and 32 features consisting of demographic data, habits, and medical history. Data was collected via questionnaires, and patients were not obliged to answer to all questions, either for privacy reasons, or for the non-pertinency of the specific question (e.g., patients with no medical history decided to skip most of the

questions related to this aspect because they simply did not apply).

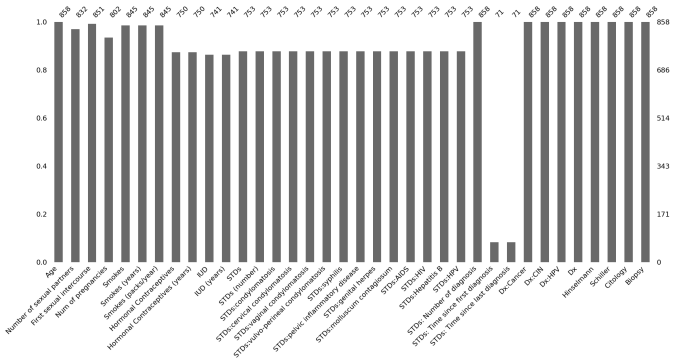


Fig. 1. Barchart displaying the amount of missing values per feature

This is one crucial aspect, as it characterizes the most important data quality issue of this dataset: a relatively high missing-data rate. The way I decided to deal with this problem led to the second goal of this research: analyzing how different data imputation methods influence the classifiers' performance at the modeling stage.

Apart from this point, which will be explained deeper in the next section, the second data quality issue of this dataset is that of high class imbalance. As it would be expected from a medical screening dataset, patients that tested negative to the four tests make up the majority of all instances, leaving very a small room for those who tested positive. In fact, all the four tests are characterized by a positive rate below 10%. The Exploratory Data Analysis (EDA) was essential to understand the data, and also the context around it. First of all, it was enough to just display the dataset to make one concern arise: the dataset had no apparent missing values, but only because all the cells that were supposed to be empty were filled with "?" characters. This first insight, other than implying that such values had to first be removed before proceeding with further analyses, also explained why all columns with such characters were assigned the "object" data type (i.e., it would have otherwise prevented the column from containing both numerical values and string values). This very first operation required some preliminary data cleaning steps to be performed, as I proceeded at substituting the "?"s with NaN values, thus allowing me to change the data type of the "object" columns. After this cleaning step, the data was suitable for being plotted and extracted further information from.

The first analysis aimed at understanding what was behind all those missing values, and how these were distributed among the features of the dataset. Both a barchart and a correlation heatmap were used for this purpose, and from these plots it was possible make some useful considerations about the data. The barchart allowed for a comprehensive view on the missingness of data: from the plot I was able to group the features according to the amount of missing data within them. Some columns, typically those referred to non-sensitive data, were completely filled up; other columns suffered from 100 missing entries more or less, but two columns in particular highly diverged from these measurements, both containing only 71 entries out of 858. These two columns referred, respectively, to the time elapsed since the first and from the last STD diagnosis. High data missingness was likely due to the fact that these two fields only applied for patients featuring STDs in their clinical history, which were a notable minority in the whole dataset. The missingness-based correlation heatmap gave out a different level of insights, as it revealed some sort of patterns in data missingness. This pattern clearly reflected the behavior of the patients when compiling the questionnaire: for example, leaving the binary field "STDs" blank meant leaving all the other STD-related fields blank. This analysis step, even though it had one other main purpose, revealed some interesting information about how the features were correlated one another even before carrying out the correlation analysis. This dataset is structured in a specific sense: there's a few variables that "anchor" some other variables. For example, from a negative answer to the "STD" question (equal to the "0" binary value) it is possible to predict with absolute certainty the values for the other 15 STD-based features: it is clear that, if one patient never contracted any STDs, all the other fields asking for specific subtypes of STDs were to be answered negatively as well. The same reasoning can be applied to explain the missingness patterns, as already mentioned.

After the reasons for missingness were cleared out, the next step in the EDA pipeline was that of identifying actual correlations between the data. From this analysis it was possible to understand how different features related with each other, and the results deriving from it confirmed the suspects arisen in the previous step.

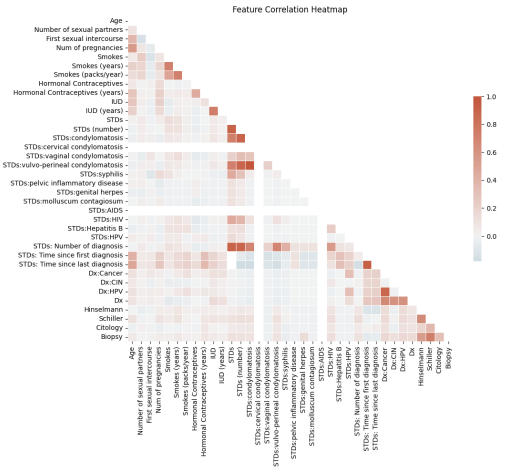


Fig. 2. Feature Correlation Matrix

Eight couples of attributes showed very high levels of redundancy (with correlation score greater than 0.80), and these columns were all referring to STDs (thus confirming the aforementioned assumptions), exception made for the two columns with high missing data rates, because their correlation was a consequence of their "time" nature. In addition to this discovery, one other discretely correlated couple turned out

to be that between the Schiller's and the Biopsy tests. This measure is not casual, but rather it depends on a purely medical fact: these tests are part of the same diagnostic funnel, and positivity to Schiller's test means that one patient must undergo biopsy, but also that this very same patient has 73% of possibility that the Biopsy test yields the same result as the previous diagnostic test. Consequently to the general correlation analysis, the focus shifted on the four target variables specifically. The first kind of analysis revolved around Class Imbalance Visualization, so as to have a graphical representation of how skewed towards the majority label was each variable's data. Results confirmed the expectations: positive rates are very low, ranging from Schiller's 8.62% to Hinselmann's 4.08%.

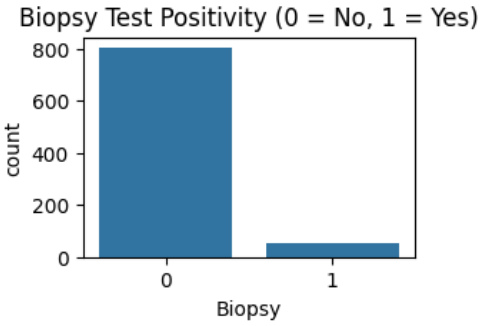


Fig. 3. Countplot showing class imbalance for the Biopsy Test

In line with what found in the previous correlation analysis, the next move was to plot a correlation heatmap that, differently from before, only displays correlations between the four targets, so to understand how these tests relate to each other. Other than the Schiller-Biopsy relation mentioned earlier, I was able to understand that, although weaker, Hinselmann's test shows some signs of correlation with these two tests too (peak at corr == 0.65 with Schiller's. As for Cytology, it is the most "set apart" target, being very weakly correlated with the other tests (peak at corr == 0.36, again with Schiller's).

Moving on from intra-target correlation analysis, but still insisting on the matter, Bivariate Analysis was performed, so to investigate on how individual features correlate with the four targets. Since doing it for all features would've been to complex to handle, I decided to select only three of them, choosing from the most common risk factors, meaning Age, years of smoking, and years of taking hormonal contraceptives.

- "Age": contrary to general cancer trends, age did not demonstrate a highly distinct separation between positive and negative instances in this specific screening cohort;
- "Smoking (years)": a prolonged smoking habit emerged as a considerable risk factor, with positive diagnostic cases clustering prominently among patients with over 10 years of smoking history;
- "Hormonal Contraceptives (years)": similarly, long-term usage of hormonal contraceptives (i.e., exceeding 10 years) showed a meaningful impact on positive diagnostic outcomes.

Finally, outlier investigation and zero-variance checks were executed to bridge the EDA and Data Preparation phases. These steps successfully identified invariant features providing no predictive value, as well as highly skewed distributions requiring targeted mathematical transformation prior to modeling.

### B. DATA PREPARATION

Thanks to EDA, the tasks to perform in the Data Preprocessing stage were made clear, and will now be discussed. This part of the study can be further split in two: one preliminary part where the dataset undergoes cleaning and transformation, and an "imputation" part, where the cleaned dataset gets filled according to two different methods: one based on the median and frequency, and the other based on KNN.

As for the cleaning part, several operations were chained one after another in a logical order:

1) Basic Cleaning: once the dataset is loaded, "?" values are turned into NaN, duplicate rows are dropped, and "object" columns are converted to a numeric data type;
2) Zero-variance drop: columns having zero variance are dropped, along with the two columns with too many missing values to be relevant;
3) High-correlation drop: selecting one feature off each of the highly correlated couples spotted during EDA and dropping them. Elimination followed one precise criterion: as all these features shared high correlation with the "STDs (number)" feature, the latter was kept as it provides all that is necessary to the classifiers to learn. In fact, these high correlation couples are not a coincidence: in the eyes of a classifying algorithm all these dropped columns do not provide further information for classification purposes.
4) Additional redundant features drop: following the same kind of reasoning as before, I also decided to drop other columns deemed to provide redundant information. For example, binary flags such as "Smokes" or "Hormonal Contraceptives" are not useful if we're tracking the relative count (e.g., Hormonal Contraceptives (years) simply yields 0 if no hormonal contraceptives have been taken at all);
5) Low-variance features aggregation: in addition to features with zero variance, the EDA revealed of several features characterized by very low variance. Since all these features referred to different types of STDs, instead of dropping them, I decided to aggregate and differentiate them in two columns: one filled with viral-based STDs, and the other with bacterial and inflammatory-based STDs.
6) Logarithmic transformation of right-skewed features: outlier investigation was crucial to identify all those features characterized by an imbalanced distribution. Logarithmic transformation was applied to these features so as to normalize their distribution and prevent them from influencing classifiers' performance.

After the last operation, results were saved into a new .csv file to be used for the second part of the Data Preprocessing pipeline: data imputation.

Due to the different nature of the two imputation methods, missing data was filled via two different pipelines. In general, the two methods followed the same reasoning: first load the data from the cleaned dataset, then perform a logical sequence of operations on the data, and finally save back the result in the .csv file associated to the imputation method. The script for KNN-imputation was written first: the preliminary operation to be executed on the data was that of data scaling, as distance-sensitive algorithm like KNN are influenced by the different scale of the data they should work on. The scaling choice that I've adopted throughout the study relied on the MinMaxScaler algorithm, as it preserves zero-values as true zeros (crucial aspect in a dataset where the majority of the features are binary), and it also provides stability to the KNN imputation algorithm. The output of this process consists in the newly scaled dataset (exception made for the targets), and a "scaler" object that makes it possible to reverse the scaling process if the next operations needed it. Imputation was then performed on the scaled data, followed by one more important operation: anomaly detection. The goal of this operation was to identify anomalous combination of values among all the instances, via Isolation Forest, by assigning a binary anomaly label and an anomaly score to each instance. The lower the anomaly score, the more likely the instance is an anomaly with respect to the "normality". This further step creates two more columns in the dataset ("anomaly-label" and "anomaly-score"), and this addition will become an important clinical hint that the algorithms will take into account when classifying. As the KNN-imputation came to an end, previously log-transformed and scaled features were brought back to normality, and discrete columns were rounded to the the nearest integer.

The second imputation workflow, based on median and frequency, was simpler to develop, as it makes use of the already-defined functions of before, and consists of three processing steps: first the original, cleaned dataset is imputed using a SimpleImputer algorithm that fills the missing numerical features with the median, and uses frequency for the remainder binary ones (excluding "Age" because it has no missing values, and the targets); secondly, imputed data is scaled so to be performed anomaly detection upon; finally, the dataset is inverse-scaled and log-transformation is reversed as well.

At last, Data Preprocessing was crucial, as it guided the original datasets through chains of transformations and manipulations that ultimately created two different datasets, ready to be fed to all the intended classifiers in the Modeling phase.

### *C. DATA MODELING*

The Data Preprocessing phase was obviously the most crucial part of the work. The result is the refined and cleaned version of the original dataset, making it possible to perform the intended classification task. In fact, the data modeling phase consisted in setting the pipeline, the sequence of classifiers to be used to against the data in order to build up a model indeed: something that understands the nature of the data and is able to make predictions out of it. As mentioned earlier, the output of the Data Preprocessing phase is double-faceted: one KNN-imputed dataset, and a median and frequency-imputed one. Both the two datasets had to undergo the same modeling pipeline, so as to ultimately understand which was the one behind the most successful classification performance.

The Modeling pipeline consists of five different classifiers, each with its own strengths and weaknesses: Logistic Regression, Random Forest, XGBoost, Support Vector Machine (SVM), and Multilabel KNN. The way the dataset was split made use of a MultiLabelStratifiedShuffleSplit cross-validator, which was perfectly suited for the task at hand. In fact, this splitting algorithm randomly divides the dataset in several folds, while ensuring that each fold always contains the same proportion of target labels. Cross validation and stratified sampling are two necessary conditions to be imposed on the splitting in order to ensure a viable classification process on such an imbalanced and multilabel dataset. In particular, since I decided to have a 80/20 split of the dataset into training and test set, the cross-validator was set to create five different folds of the data, so to make the classifiers see all the data once, at least.

Considering all that was said until now, each classifier is trained and tested on five different folds of the main dataset, and its performance on each fold is saved into a report in terms of precision, recall, and f1-score. Once the cycle ends, the five different scores for each metric are averaged, and finally displayed as a Pandas DataFrame at the end of each classifier's run.

## III. RESULTS

Prior to the actual classification, two thresholds of acceptability for the precision and the recall metrics were established. The choice for the thresholds was the result of a compromise between the medical and the statistical significance: in particular, I set a threshold of 0.6 for the recall, and 0.2 for the precision. A high recall is prioritized in the medical domain, as it is wanted not to misclassify any True Positive. However, such an high score for the recall is the result of a huge sacrifice in terms of precision. Since the goal was not that of blindly label of patients as positive and making them go under stressful and potentially useless tests, a 0.2 threshold for the precision was deemed to be the best solution to balance the needs for a high recall rate, a less stressful diagnosis process, and statistical significance.

For all these reasons, and also because of its diagnostic importance, performance on the Biopsy test was selected as the primary evaluation criterion to determine the best classifier. In addition to each test's metric scores I decided to also track the Macro Average for all tests: this measure tells how well the classifier performs across all four medical targets, considered equally.

After having trained and tested the various models in both the two differently-imputed versions of the same dataset,

it was time to evaluate the performance of each classifier, comparing it against that of the other counterparts in the two versions. Splitting the modeling phase in two separate ”storylines” proved to be an insightful initiative: the KNN-based imputation strategy turned out to be, as expected, the winning choice. This serves to achieves one of the two primary objectives for this study: the choice for the right imputation method has critical importance for the future development of the ML pipeline. Different contexts require different specific choices to be made, and in this case KNN was the right choice because of its inherent capability of exploring similarities between various patients in order to estimate plausible values. Contrarily, the median-based imputation strategy proved well its non-suitability for the specific context of analysis: the feature space simply collapsed, as every single classifier failed to meet the precision target. Linear and tree-based models suffered tremendously from such an imbalanced dataset: Logistic Regression did not even meet the recall target, while XGBoost and Random Forest failed to find clean mathematical boundaries and drastically dropped their decision thresholds, creating a general rule that seemed to maximize the recall at all costs. KNN and SVC (Support Vector Classifier) performed slightly better, but still failed to reach adequate levels of precision. Median imputation negatively affected their data-map, making it harder for these two distance-based classifiers to distinguish one cluster of instances from another.

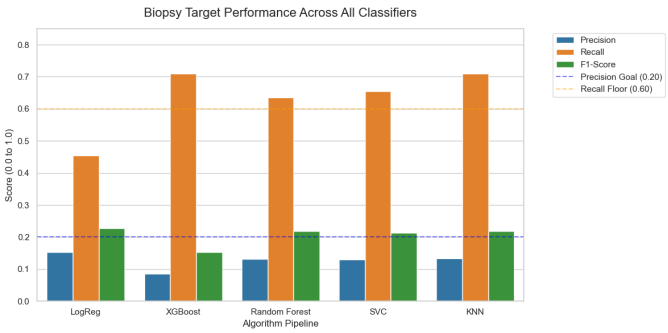


Fig. 4. Classification results for Biopsy on median and frequency-imputed dataset

After having proved that KNN was indeed the right imputation strategy between the two, it was time to deeply analyze how well classifiers performed in this ”winning” scenario.

Again, Logistic Regression was the one doing worst overall, despite performing better than all the classifiers of the previous scenario (consider this as a further confirmation of median-imputation inadequacy), it still missed the precision target and barely hung on the minimum acceptable recall for the Biopsy test. This classifier is simply too rigid to capture the complex, non-linear medical profiles of these patients.

Random Forest proved to be a non-ideal classifier too. With a 0.074 score for precision, and 0.818 score for recall, this classifier performed even worse than before. The goal was basically not to miss a single True Positive, but this intention resulted in unacceptable precision levels. Overly imbalanced targets like Hinselmann and Cytology yielded even higher recall score, peaking at 1.00 for Cytology. The algorithm simply surrendered and decided to flag as positive minimally borderline patients too.

XGBoost, instead, showed that embedding a boosting strategy to a collection of Decision Trees is an optimal solution for this specific dataset. KNN-imputed data clusters were more clearly defined, hence this algorithm was not only able to infer clearer choices for the split points, but also its iterative nature allowed for further refinements with each tree being built after the other. Its efficiency is testified by its scores for the Biopsy test: 0.2 for precision (the exact threshold), and 0.655 for recall.

SVC achieved the same score as XGBoost for the recall, while being slightly less precise (score: 0.196). Using a non-linear kernel as a hyperparameter (i.e., a polynomial kernel), SVC is able to project data into a higher dimension to find non-linear boundaries between the points. What made the difference between the two models, in this specific scenario, is that SVC ultimately creates a smooth geometric shape that globally applies to the dataset, eventually missing granular combinations of values. Furthermore, this algorithm takes into account all features when setting the decision boundaries, while XGBoost simply does not use features that do not improve the Information Gain.

Multilabel KNN turned out to be the best classifier overall, balancing an acceptable recall score (0.655) with the highest scored ever achieved for precision in this study (0.270). This outcome was a quite predictable one: having shaped the original dataset with an imputation method based on the same algorithm, it was possible for the KNN classifier to operate on a ”familiar” spatial data map. The imputer pre-clustered the patients based on clinical distance, and the classifier perfectly read those distance-based clusters: that’s what is called ”algorithmic synergy”.

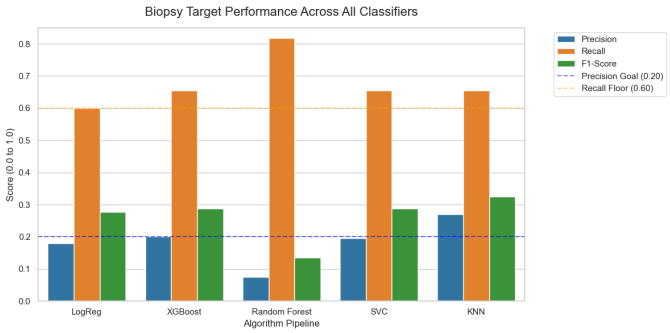


Fig. 5. Classification results for Biopsy on KNN-imputed dataset

While the models met the predefined thresholds, the outcomes reinforce already existing doubts regarding the data itself and, at the same time, it testifies that there is plenty of room to improve measurements.

Having set the classifiers so that they always give priority to recall, these models have all opted to sacrifice precision for the sake of not missing as many True Positive patients as possible.

Even though this means that it is unlikely that one patient is wrongly flagged as negative, the specific high recall-low precision tradeoff also implies that lots of patients are wrongly classified as positive. This choice causes an increased overload both in psychological terms for the False Positive patients, and in workload terms as it concerns the medical personnel.

## IV. Conclusions

This study aimed at constructing a complete Machine Learning pipeline, following the guidelines and principles of the CRISP-DM methodology. The primary objective was to perform a multilabel classification task for the cervical cancer diagnostic funnel, with the view of moving beyond the prediction of a single outcome, but also to understand how multiple variables (i.e., risk factors) relate to each other with respect to a diagnostic outcome.

As the cervical cancer dataset was characterized by highly sparse features and missing values, crucial operations of the Data Preparation step were data cleaning and feature aggregation. However, among all the necessary and equally important tasks, the decisive step would have been the choice for the imputation strategy to adopt to fill the missing entries. Among the two candidate methods, median and KNN imputation, the latter proved superior because it leveraged the similarities between patients' clinical profiles: this was an essential need in a medical dataset with complex, non-linear relationships.

The actual comparison between the two strategies was carried out in a Modeling stage, where several different classifiers, Logistic Regression, Random Forest, XGBoost, SVM, and KNN, were trained and tested against the cleaned dataset. Given the severe class imbalance affecting the data, stratified sampling and cross-validation were the choices made at validation time, so as to help classification algorithms distinguish between negative and positive patients and generalize to unseen data without overfitting. Furthermore, as the context is that of medical diagnosis, the leading decision was that of creating "conservative" models that prioritized high scores for recall, at the cost of sacrificing precision. In other words, it was more important to avoid False Negatives (missing a sick patient) than to avoid False Positives (wrongly classifying a healthy patient), as cancer screening demands.

At the end of the process, the best performing model was the KNN classifier, which fully exploited the conformation of the KNN-imputed dataset. This classifier created a model that satisfied both the previously set target scores for precision and recall, showing higher understanding of the various non-linear relationships clustering different patients together.

Overall, the results were satisfactory, at least from an academic perspective. The relatively small size of the dataset, combined with an extreme class imbalance, put a hard ceiling on how much a traditional ML model can learn without overfitting. In any case, there remains substantial room for improvement. First of all, adding a deep learning architecture, or a neural network to the classification pipeline is a logical next step. However, while conducting the study, many other different opportunities arose, such as applying advanced synthetic data generation (e.g., SMOTE) to better balance the dataset and increase classifiers' abilities, or further exploiting the relationships between the four diagnostic tests (i.e., using Schiller's test outcome as an additional feature to predict the Biopsy test's).

Ultimately, while this study serves as a foundation for future, more advanced studies, this study reinforces a fundamental principle: high-quality data yields high-quality models. Every single decision that is taken over the course of a ML study inevitably impacts the final result, and it is on the final result that higher-level considerations are derived. Cancer screening demands significantly more reliable models, but still this very same work demonstrates how Artificial Intelligence will better support medical screening in the near future.